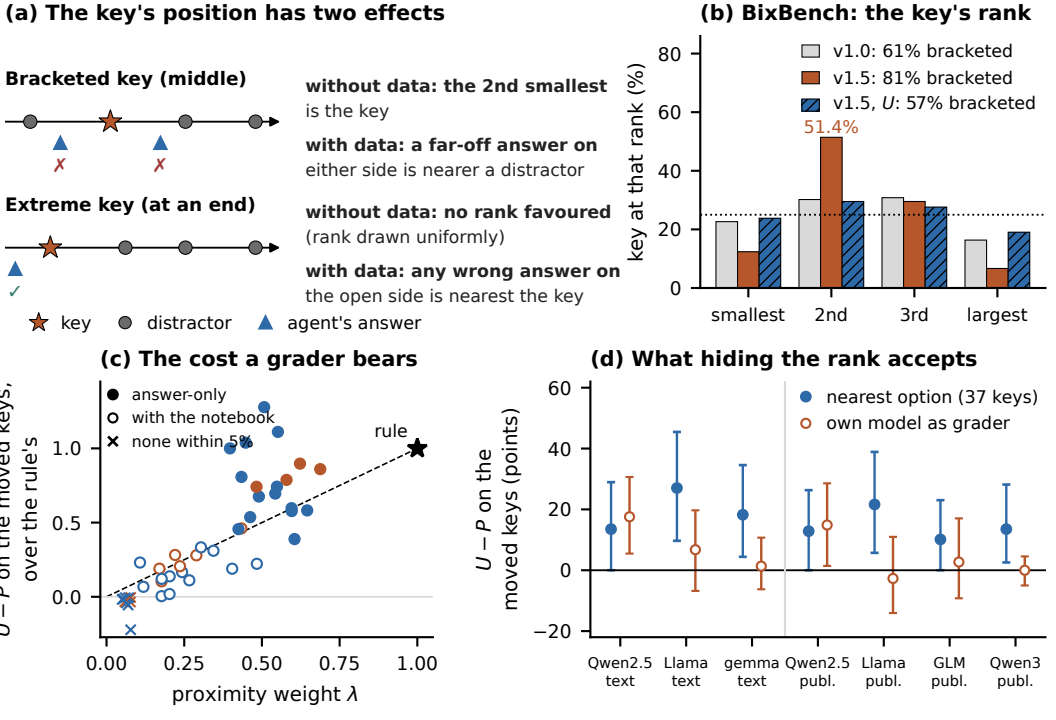
(a) The key's position has two effects
Bracketed key (middle)
without data: the 2nd smallest is the key
with data: a far-off answer on either side is nearer a distractor
Extreme key (at an end)
without data: no rank favoured (rank drawn uniformly)
with data: any wrong answer on the open side is nearest the key
key
distractor
agent's answer
(b) BixBench: the key's rank
v1.0: 61% bracketed
v1.5: 81% bracketed
v1.5, U: 57% bracketed
51.4%
key at that rank (%)
smallest
2nd
3rd
largest
(c) The cost a grader bears
answer-only
with the notebook
none within 5%
rule
U − P on the moved keys, over the rule's
proximity weight λ
(d) What hiding the rank accepts
nearest option (37 keys)
own model as grader
U − P on the moved keys (points)
Qwen2.5 text
Llama text
gemma text
Qwen2.5 publ.
Llama publ.
GLM publ.
Qwen3 publ.